Numerically Solving the Forced van der Pol Equation

A Finite-Difference Study with Applications to Economic Dynamics

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The forced van der Pol equation is a standard reduced form of Kaldor's and Goodwin's nonlinear business-cycle models. This note solves it with a second-order finite-difference scheme and studies how the strength of the nonlinearity and the amplitude of an external periodic driver shape its long-run behavior, using stroboscopic Poincaré sections and largest Lyapunov exponents to classify each case. Most settings are regular: weakly nonlinear or weakly forced systems oscillate quasi-periodically, and strongly nonlinear or strongly forced systems lock onto the forcing. The case study is chaotic, but its forcing amplitude lies within a millionth of the point at which its chaotic attractor is destroyed, so close that the finite-difference step size decides whether the computed solution eventually settles into a regular cycle.

Written in August 2024; revised in October 2026. The revision corrects the numerical scheme, redraws every figure, adds stroboscopic Poincaré sections and Lyapunov exponents to classify each case, and corrects several statements about the dynamics that those diagnostics contradict. The revision notes list what changed.

1 Introduction

1.1 Complex dynamics in economics

A dynamical system describes how a state evolves over time. Following Day (1994), as summarized by Rosser (1999), a deterministic system is complex if it does not settle down to a fixed point, a limit cycle, or explosive growth. The Santa Fe view of complexity adds features of the economy as a whole, including dispersed interaction among heterogeneous agents, the absence of a global controller, continual adaptation, perpetual novelty, and out-of-equilibrium dynamics Rosser (1999). Nonlinear models are the analytical core of this program, and they have been used to study business cycles, growth, price dynamics, and financial markets (Sportelli and Mastrorillo 2014). Their long-run behavior can be a steady state, a periodic cycle, quasi-periodic motion combining two incommensurate frequencies, or chaos, in which nearby trajectories separate exponentially and long-run prediction fails (Strogatz 2024).

Two classic nonlinear models of the business cycle, Kaldor's investment–saving model and Goodwin's nonlinear accelerator, reduce to forms of the van der Pol equation. Adding a periodic exogenous driver gives the forced van der Pol equation, which can show all of these behaviors (Chian 2007, 9).

1.2 Kaldor's model

Kaldor (1940), building on Keynes (1936) and on Kalecki's investment model (Kalecki 1935), proposed a business cycle driven endogenously by the interplay of investment and saving. Following Chang and Smyth (1971), and taking saving to depend on output alone, Kaldor's model can be written as a pair of differential equations for output Y and the capital stock K,

$$\dot{Y} = \alpha[I(Y,K) - S(Y)], \qquad \dot{K} = I(Y,K), \tag{1}$$

where $\alpha > 0$ is the speed at which output responds to excess demand. Differentiating the first equation, using the second, and substituting $I = S + \dot{Y}/\alpha$ gives

$$\ddot{Y} = \alpha(I_Y - S_Y)\,\dot{Y} + \alpha I_K\, I = [\alpha(I_Y - S_Y) + I_K]\dot{Y} + \alpha I_K\, S(Y).$$

If investment is linear in the capital stock, so that I_K is a negative constant, this is a Liénard equation $\ddot{x} + A(x)\,\dot{x} + B(x) = 0$, with output measured relative to its equilibrium,

$$A(Y) = -[\alpha(I_Y - S_Y) + I_K], \qquad B(Y) = -\alpha I_K\, S(Y).$$

Kaldor's central assumption is that investment responds to output more strongly than saving near normal activity and less strongly at extreme levels. That makes the equilibrium unstable while keeping fluctuations bounded: in the Liénard form, $A < 0$ near equilibrium and $A > 0$ far from it. The simplest symmetric function with this sign pattern is $A(x) = \mu(x^2 - 1)$, and with $B(x) = x$, which corresponds to linear saving after rescaling, the model becomes the van der Pol equation (Chian 2007, 12–13):

$$\ddot{x} + \mu(x^2 - 1)\,\dot{x} + x = 0. \tag{2}$$

The cubic form is a modeling choice that captures Kaldor's assumption, not something derived from it.

1.3 Goodwin's model and periodic forcing

Goodwin (1951) proposed a nonlinear accelerator–multiplier model as an alternative to the linear models of the time, whose cycles either died out or exploded depending on parameter values. His model can be written

$$\varepsilon\theta\,\ddot{y} + [\varepsilon + (1-\alpha)\theta]\dot{y} - \varphi(\dot{y}) + (1-\alpha)\,y = I(t),$$

where y is income, α the marginal propensity to consume, ε and θ lags, $\varphi(\dot{y})$ the investment induced by changes in income, and $I(t)$ autonomous investment. Because the nonlinearity acts on the rate of change of income rather than on its level, the equation is of Rayleigh type, $\ddot{z} + F(\dot{z}) + z = f(t)$ after rescaling. Differentiating a Rayleigh equation and setting $x = \dot{z}$ turns it into a van der Pol equation, with the forcing replaced by its derivative; a sinusoidal forcing stays sinusoidal at the same frequency. Lorenz (1987) showed that when autonomous investment varies periodically, Goodwin's model can generate chaotic motion. Following Chian (2007, 14), we study the forced van der Pol equation

$$\ddot{x} + \mu(x^2 - 1)\,\dot{x} + x = a\sin(\omega t). \tag{3}$$

1.4 The forced van der Pol equation

Consider first the unforced equation Equation 2. Its middle term is a damping term whose sign depends on the state. When $|x| > 1$ it removes energy and shrinks the oscillation; when $|x| < 1$ it adds energy and amplifies it. Small oscillations therefore grow, large ones shrink, and trajectories approach a closed orbit on which the energy gained near the origin and lost away from it balance over each cycle (Strogatz 2024, sec. 7.1). By Liénard's theorem this limit cycle exists and is unique and stable for every $\mu > 0$ (Strogatz 2024, sec. 7.4). In general a stable limit cycle attracts only the trajectories that start in its basin; for the van der Pol equation the basin is every initial condition except the equilibrium at the origin.

The cycle's shape depends on μ. As $\mu \to 0$ the damping term vanishes, the cycle approaches a circle of radius 2, and the motion becomes nearly sinusoidal. For $\mu \gg 1$ it becomes a relaxation oscillation: slow drifts along two branches of a cubic curve, interrupted by fast jumps between them (Strogatz 2024, secs. 7.5–7.6). At $\mu = 0$ the equation is the harmonic oscillator. A cycle of amplitude near 2 exists on both sides of that point, stable for $\mu > 0$ and unstable for $\mu < 0$, so it does not grow out of the equilibrium as a standard Hopf bifurcation would; the bifurcation at $\mu = 0$ is degenerate (Strogatz 2024, sec. 8.2). The unforced equation, being autonomous and planar, cannot be chaotic.

The forcing changes this. With $a\sin(\omega t)$ on the right-hand side, the state is $(x, \dot{x})$ together with the phase of the forcing, ωt modulo 2π, so the phase space is three-dimensional. The Poincaré–Bendixson theorem, which rules out chaos in planar autonomous systems (Strogatz 2024, sec. 7.3), no longer applies. The natural way to look at the forced system is through its stroboscopic map, which records $(x, \dot{x})$ once per forcing period $T = 2\pi/\omega$. A solution locked to the forcing appears as a finite set of points, a quasi-periodic solution as a closed curve, and a chaotic solution as a more complicated, typically fractal set.

Two further properties matter for interpretation. First, Equation 3 is unchanged if x is replaced by $-x$ and time is shifted by half a forcing period, so its cycles are symmetric: expansions and contractions mirror each other. On its own the model therefore cannot produce the asymmetry of observed business cycles, in which downturns are typically sharper than upturns. Second, the parameters have natural economic readings. Through Kaldor's reduction, μ measures how strongly the investment–saving nonlinearity pushes output away from equilibrium when it is close and pulls it back when it is far. The forcing amplitude a and frequency ω describe an exogenous driver, such as periodically varying autonomous investment or the influence of the world economy on a national economy (Chian et al. 2005). These readings are analogies, not implications of the model.

2 Method

2.1 Finite differences

The finite-difference method replaces derivatives with differences on a grid of time points. For a smooth function f and a step Δt, Taylor's theorem gives

$$f(t + \Delta t) = f(t) + \Delta t\, f'(t) + \tfrac{1}{2}\Delta t^2 f''(t) + \tfrac{1}{6}\Delta t^3 f'''(t) + O(\Delta t^4),$$
$$f(t - \Delta t) = f(t) - \Delta t\, f'(t) + \tfrac{1}{2}\Delta t^2 f''(t) - \tfrac{1}{6}\Delta t^3 f'''(t) + O(\Delta t^4).$$

Keeping the first two terms of either expansion gives the forward and backward differences, which are accurate to $O(\Delta t)$. Subtracting the expansions cancels the even-order terms and gives the central difference for the first derivative; adding them cancels the odd-order terms and gives the central difference for the second derivative:

$$f'(t) = \frac{f(t+\Delta t) - f(t-\Delta t)}{2\Delta t} + O(\Delta t^2), \qquad f''(t) = \frac{f(t+\Delta t) - 2f(t) + f(t-\Delta t)}{\Delta t^2} + O(\Delta t^2). \tag{4}$$

2.2 A scheme for the forced equation

Write X_n for the approximation to $x(t_n)$ at $t_n = n\Delta t$. Replacing the derivatives in Equation 3 by the central differences Equation 4 at t_n gives

$$\frac{X_{n+1} - 2X_n + X_{n-1}}{\Delta t^2} + \mu(X_n^2 - 1)\frac{X_{n+1} - X_{n-1}}{2\Delta t} + X_n = a\sin(\omega t_n),$$

and solving for X_{n+1},

$$X_{n+1} = \frac{\Delta t^2(a\sin(\omega t_n) - X_n) + X_{n-1}(k_n - 1) + 2X_n}{1 + k_n}, \qquad k_n = \tfrac{1}{2}\,\mu(X_n^2 - 1)\Delta t. \tag{5}$$

The scheme needs two starting values. We set $X_0 = x(0)$ and $X_1 = x(0) + \Delta t\,\dot{x}(0) + \frac{1}{2}\Delta t^2\ddot{x}(0)$, with $\ddot{x}(0)$ from Equation 3, and estimate the velocity by the central difference $V_n = (X_{n+1} - X_{n-1})/2\Delta t$. As in 2024, every run starts from rest, $x(0) = \dot{x}(0) = 0$, with $\Delta t = 0.01$ unless stated otherwise.

2.3 Checking the scheme

Table 1: Error of the finite-difference solution at $t = 20$ for the case study, against a fourth-order Runge–Kutta solution with 2,000,000 steps. The order is $\log_2$ of the ratio of successive errors.

Step	Error, forcing at t_n	Order	Error, forcing at t_{n-1}	Order
0.0400	2.45e-03		2.70e-02	
0.0200	6.12e-04	2.00	1.42e-02	0.93
0.0100	1.53e-04	2.00	7.27e-03	0.96
0.0050	3.83e-05	2.00	3.68e-03	0.98
0.0025	9.56e-06	2.00	1.85e-03	0.99

Table 1 checks the scheme against a fourth-order Runge–Kutta (RK4) solution computed with a much smaller step. With the forcing evaluated at t_n, the error falls by a factor of four each time the step halves, as a second-order scheme should. The 2024 version evaluated the forcing at t_{n-1}, which shifts its phase by $\omega\Delta t$ and reduces the scheme to first order: its errors only halve.

This is the right kind of check for a chaotic system. Trajectories computed with different step sizes stay close only for a while, because small differences grow exponentially, so agreement over long horizons is not to be expected. What should agree over long horizons are summaries of the long-run behavior, which the next section introduces.

2.4 Diagnostics for long-run behavior

Two diagnostics classify the long-run behavior. The stroboscopic Poincaré section plots $(x, \dot{x})$ once per forcing period after transients have died out (Section 1.4). The largest Lyapunov exponent λ measures the average exponential rate at which nearby trajectories separate. We compute it by integrating the variational equation of Equation 3 alongside the trajectory, renormalizing the tangent vector once per forcing period, and averaging the logarithmic growth (Benettin et al. 1980; see also Wolf et al. 1985). A positive exponent indicates chaos, a zero exponent quasi-periodic motion, and a negative exponent a periodic solution.

These long runs use RK4 with 1,000 steps per forcing period ($\Delta t \approx 0.014$), a fourth-order method to which the finite-difference scheme converges. Each exponent is computed from three starting points, discarding 2,000 forcing periods and averaging over the next 3,000. The integrators are written in C++ and called from R through Rcpp; the code is in `vdp.cpp`, alongside this note.

3 Results

3.1 The case study

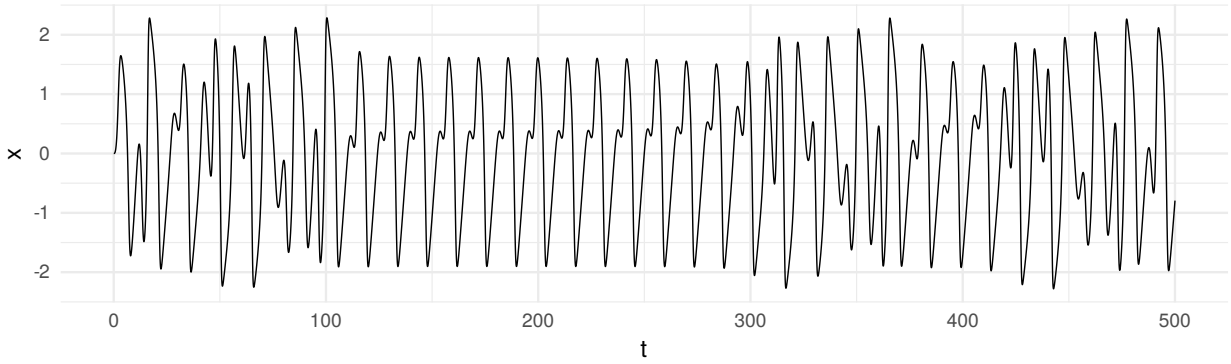


Figure 1: The case study ($\mu = 1$, $a = 0.983139$, $\omega = 0.45$), computed with the finite-difference scheme from rest.

Figure 1 shows the case study over the same horizon as in 2024. The oscillation is irregular at first, settles into a long, nearly periodic stretch, and then becomes irregular again. Figure 2 shows the same run in the phase plane and the stroboscopic section of its long-run behavior. The section is neither a finite set of points nor a smooth closed curve: its points lie near a curve but fill it unevenly, with gaps and clusters. The largest Lyapunov exponent is positive, 0.030, with the same value to within 0.0009 from all three starting points. The case study is chaotic.

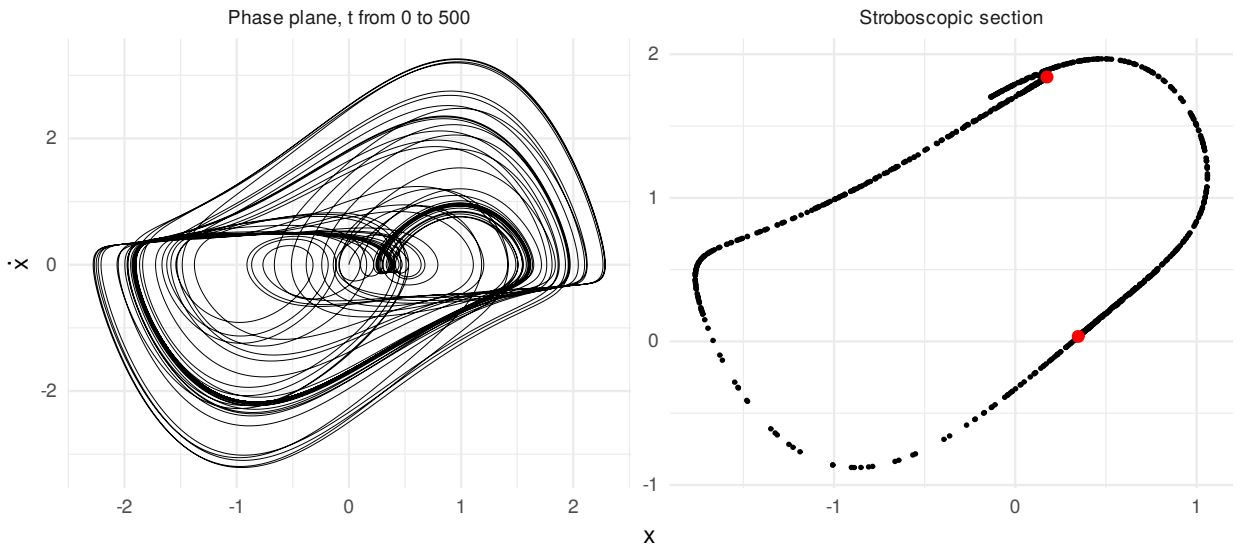


Figure 2: Left: the run of Figure 1 in the phase plane. Right: the stroboscopic Poincaré section of its long-run behavior (RK4, forcing periods 2,000 to 3,000 from rest). The two red points are the stable period-1 cycle and its mirror image, which coexist with the chaotic attractor.

The chaos sits at the edge of a window of regular behavior. Chian et al. (2005) show that at $\mu = 1$ and $\omega = 0.45$, increasing a opens a periodic window with a saddle-node bifurcation, which they place at $a = 0.98312$. Following the period-1 cycle born there with Newton's method locates the bifurcation at $a = 0.983118$, in agreement with their value. At the case-study amplitude the cycle already exists and is stable, with Floquet multipliers 0.62 and 0.39, but trajectories do not find it. From six different starting points, none of the runs at $a = 0.983139$ settled on the cycle within 20,000 forcing periods, whereas at $a = 0.983140$ all six settled within 878 periods. Just above the bifurcation the stable cycle and the chaotic attractor therefore coexist, and the chaotic attractor is destroyed between $a = 0.983139$ and $a = 0.983140$, consistent with a boundary crisis. The case-study amplitude sits within 10^{-6} of that boundary, on the chaotic side.[1]

The nearly periodic stretch in Figure 1 is the chaotic trajectory passing close to the period-1 cycles. In the accurate run from rest, 21% of forcing periods begin within 0.05 of one of the two cycle points in the section, in runs of up to 20 consecutive periods.

[1]Strictly, the runs show that at $a = 0.983139$ the chaotic motion persists for at least 20,000 forcing periods from all six starting points. It is either an attractor or a transient far longer than any run here.

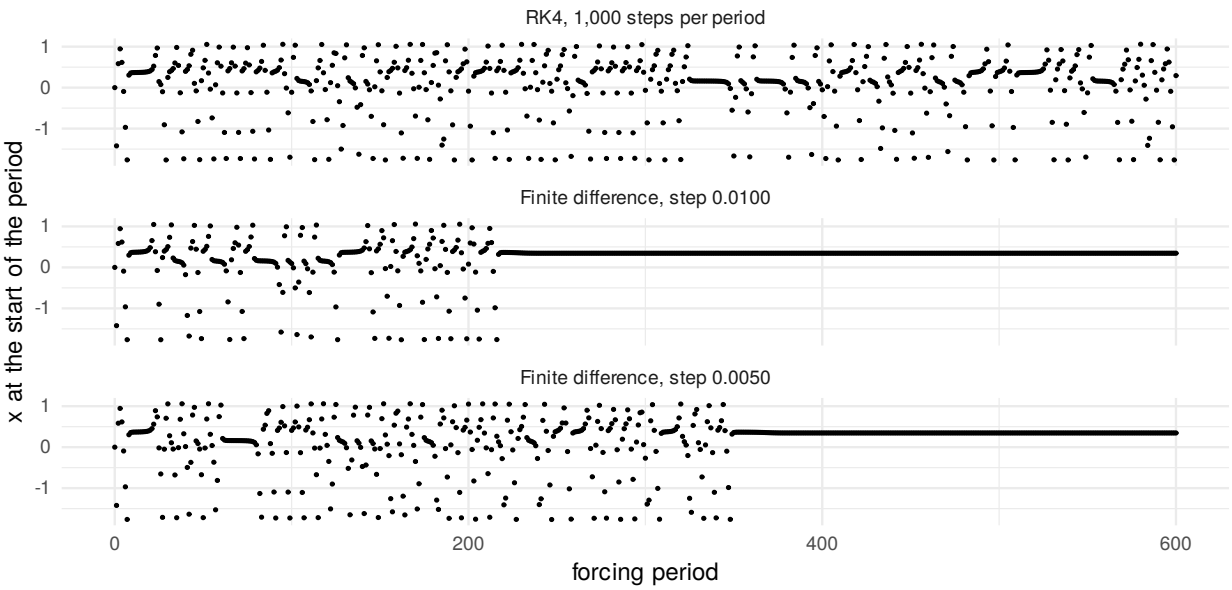


Figure 3: The case study over the long run: x sampled once per forcing period, from rest. The accurate RK4 solution stays chaotic; the finite-difference scheme settles on the period-1 cycle at a time that depends on its step.

Because the case study sits so close to the boundary, the step size of the scheme decides its long-run behavior (Figure 3). With steps of about 0.01 and 0.005, the finite-difference solution from rest settles on the period-1 cycle after 249 and 391 forcing periods. With a step of about 0.0025 it does not settle within 6,000 periods, and neither does the RK4 solution.[2] The 2024 version's figures, which stopped at $t = 500$, show only the chaotic phase that precedes the settling. Its observation that halving the step did not change the waveform holds over that horizon, but not over the long run.

Read as a business-cycle model, the case study is an economy whose exogenous driver sits at the edge of a window of regular cycles. It fluctuates irregularly, with long episodes of nearly regular cycles, while a regular cycle exists nearby that the economy, from these starting points, never settles into. The 2024 version described the time series as an economy moving between disequilibrium and equilibrium regimes. The diagnostics show instead that these regimes are phases of a single chaotic motion.

[2]These runs use steps of $T/1400$, $T/2800$, and $T/5600$, so that samples fall exactly once per forcing period. The 2024 forcing index, t_{n-1}, gives the same settling periods to within one period, as expected, since it only shifts the phase of the forcing.

3.2 Varying the nonlinearity

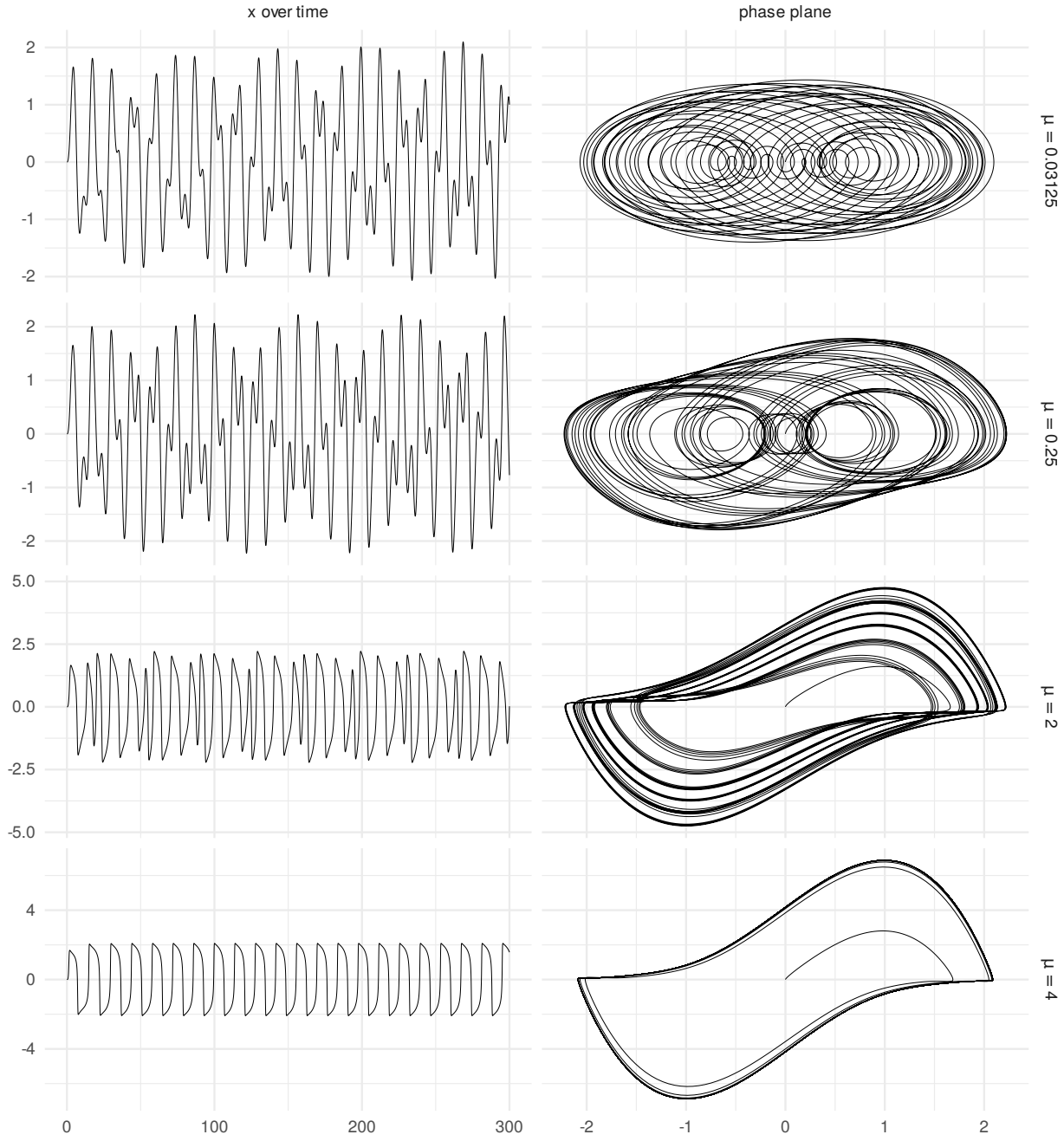


Figure 4: Varying μ with $a = 0.983139$ and $\omega = 0.45$, computed with the finite-difference scheme from rest for t from 0 to 300. Left: x over time. Right: the same runs in the phase plane.

For small μ the oscillation is quasi-periodic, not chaotic (Figure 4 and Table 2). With weak damping the system carries two oscillations at once: its own, at a frequency close to 1, and the forced response at frequency 0.45. A first-order averaging calculation makes this concrete. The forced response has amplitude $a/(1-\omega^2) \approx 1.23$, and the free oscillation survives with amplitude

r satisfying $r^2 = 4(1 - \frac{1}{2}[a/(1-\omega^2)]^2)$, so $r \approx 0.98$. Their sum explains the range of x, about ± 2.2. Because the two frequencies are not locked, the oscillation beats irregularly in time, but its Poincaré section is a closed curve and its Lyapunov exponent is zero. At $\mu = 2$ the motion is still quasi-periodic; at $\mu = 4$ the relaxation oscillation locks onto the forcing and repeats once per forcing period. Of the five values of μ studied, only $\mu = 1$ is chaotic.

3.3 Varying the forcing amplitude

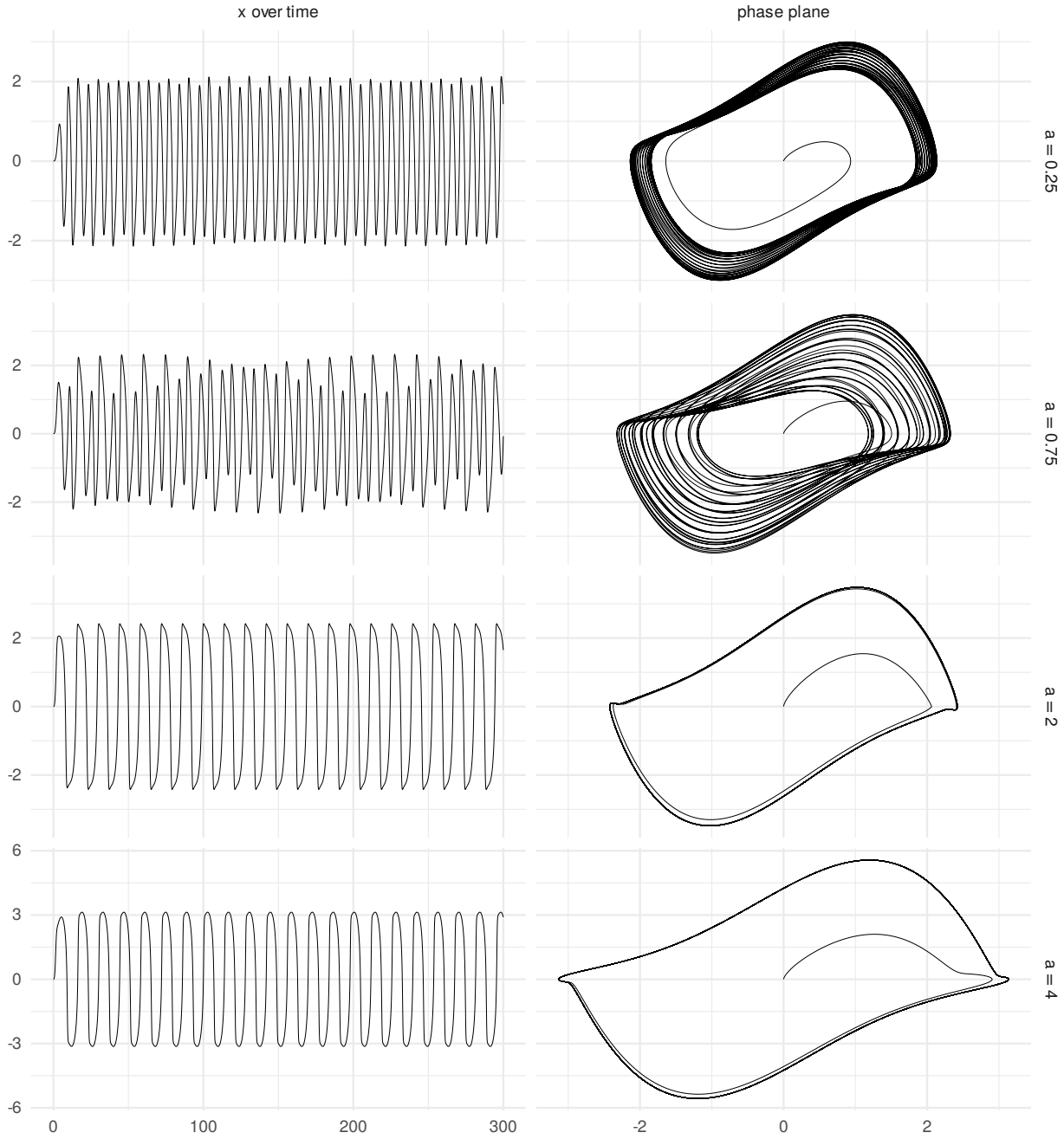


Figure 5: Varying a with $\mu = 1$ and $\omega = 0.45$, computed with the finite-difference scheme from rest for t from 0 to 300. Left: x over time. Right: the same runs in the phase plane.

Weak forcing ($a = 0.25$ and 0.75) leaves the system's own oscillation in place, and the two frequencies combine quasi-periodically (Figure 5). Strong forcing ($a = 2$ and 4) entrains the system: it oscillates at the forcing frequency, with a larger amplitude, and its Lyapunov exponent is clearly negative.

3.4 Summary of the diagnostics

Table 2: Largest Lyapunov exponents (mean over three starting points, with the range across them) and the number of distinct points in the Poincaré section (1,001 sampled periods, rounded to five decimals).

μ	a	λ	Range	Section points	Behavior
1	0.983139	0.0298	0.0009	1000	chaotic
0.03125	0.983139	0.0000	0.0000	1001	quasi-periodic
0.25	0.983139	0.0000	0.0000	1001	quasi-periodic
2	0.983139	0.0000	0.0000	1001	quasi-periodic
4	0.983139	-0.1038	0.0000	1	periodic (period 1)
1	0.25	0.0000	0.0000	1001	quasi-periodic
1	0.75	0.0000	0.0000	1001	quasi-periodic
1	2	-0.1905	0.0000	1	periodic (period 1)
1	4	-0.7189	0.0000	1	periodic (period 1)

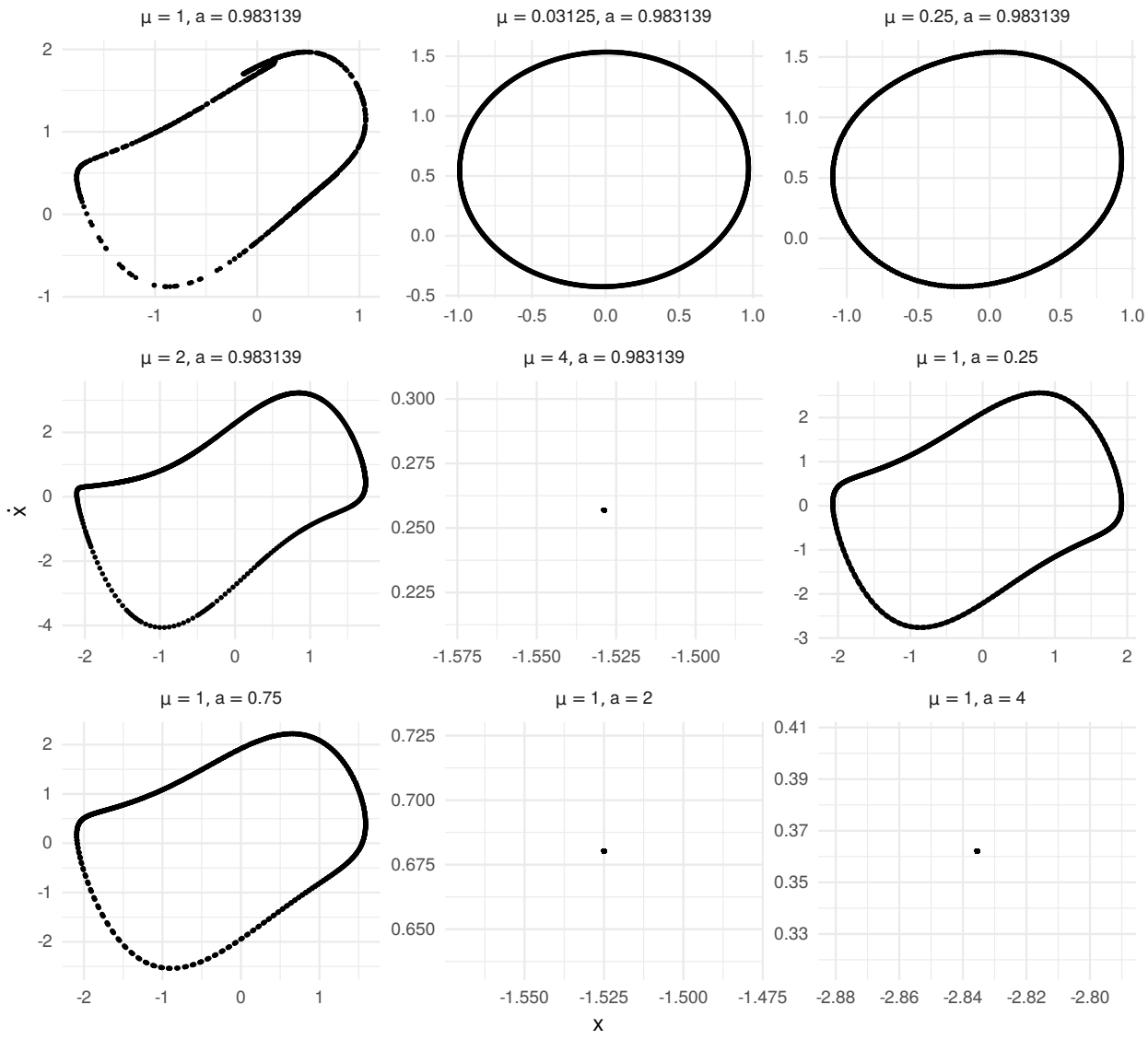


Figure 6: Stroboscopic Poincaré sections for all nine settings (RK4, forcing periods 2,000 to 3,000 from rest). Periodic solutions appear as single points, quasi-periodic solutions as closed curves, and the chaotic case study as an unevenly filled, curve-like set.

Table 2 and Figure 6 summarize the nine settings. Two patterns stand out. Complexity is not monotone in μ: chaos appears at $\mu = 1$, with quasi-periodic motion on either side and locking at $\mu = 4$. And increasing the forcing amplitude moves the system from quasi-periodic motion to locking; of the five amplitudes studied at $\mu = 1$, only the case study's is chaotic.

4 Conclusion

The forced van der Pol equation, a reduced form of Kaldor's and Goodwin's business-cycle models, can be solved accurately with a simple second-order finite-difference scheme, provided the forcing

is evaluated at the right time step. Classifying its long-run behavior needs more than a time series, however. Quasi-periodic motion looks irregular in a time series, and chaos can hide inside long, nearly regular stretches. Stroboscopic sections and Lyapunov exponents separate the cases. In the settings studied here they show that most behavior is regular, that chaos appears near $\mu = 1$ rather than at weak nonlinearity, and that the case study's chaos sits so close to the edge of a periodic window that the numerical step size decides whether a computed solution eventually settles into a regular cycle.

For economics, the lesson is about what the model can and cannot say. Read by analogy, a stronger investment–saving nonlinearity does not simply make an economy more or less stable, and a stronger exogenous driver tends to entrain the cycle rather than disrupt it. The model's symmetry also means that it cannot by itself reproduce the asymmetry of real business cycles. A natural extension replaces the periodic driver with random shocks, turning the model into a stochastic differential equation whose density obeys a Fokker–Planck equation (Saporito 2026). Recent work studies noise-driven economic cycles (Nagel et al. 2025) and noise-driven van der Pol dynamics (Yuan and Zhou 2026).

Revision notes

This note was written in August 2024 and revised in October 2026.

The numerical method was corrected. The 2024 scheme evaluated the forcing at t_{n-1} rather than t_n, which made it first-order rather than second-order (Table 1). The 2024 phase portraits appear to have plotted velocity against acceleration, since their axis ranges match $\dot{x}$ and $\ddot{x}$ rather than x and $\dot{x}$. Every figure has been regenerated from new code, and the phase portraits now show $(x, \dot{x})$.

Several statements about the dynamics were wrong and have been corrected. The descriptions of the two regimes of μ were reversed: the system approaches a harmonic oscillator as $\mu \to 0$ and shows relaxation oscillations for $\mu \gg 1$, not the other way around. The bifurcation at $\mu = 0$ is a degenerate Hopf bifurcation, and it does not generate chaos, which the unforced planar system cannot have. The 2024 version claimed that any nonlinear two-dimensional system has a unique limit cycle; this is false, and the forced system is three-dimensional in any case, so its "closed, unique limit cycle" was a projection of a chaotic attractor. The introduction described a as the strength of the nonlinear damping; it is the amplitude of the forcing.

The 2024 version classified behavior by eye from time series and phase portraits, and the new diagnostics overturn several of those classifications. The cases with small μ (0.25 and 0.03125) and with weak forcing ($a = 0.25$ and 0.75) were described as chaotic or unstable, and the case $\mu = 2$ as periodic; all five are quasi-periodic. The case study is chaotic, but the alternation the 2024 version read as regime changes is a single chaotic motion passing near a coexisting cycle, at an amplitude so close to the edge of a periodic window that the long-run outcome of the computation depends on the step size. The conclusion that complexity falls steadily as μ rises has been replaced. The appendix, which compared waveforms at two step sizes, has been replaced by the order-of-accuracy check and the long-run comparison in Section 3.1.

The economics was also corrected. The derivation of the Liénard form of Kaldor's model omitted a term, $I_K \dot{Y}$, which is now included, with the assumptions under which the van der Pol form follows. Goodwin's model is of Rayleigh type and reaches the van der Pol form by differentiation, rather

than being derived as a forced van der Pol equation directly. The note now also points out that the model's symmetry rules out asymmetric cycles, and it presents its economic readings of the parameters as analogies.

Finally, passages that followed Strogatz's wording closely have been rewritten and attributed, and the reference list was corrected: Chang and Smyth's names were in reverse order, Kalecki's paper is the 1935 *Econometrica* article, the overview of nonlinear economic dynamics is by Sportelli and Mastrorillo, and Strogatz is now cited as one edition, by section.

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